



Kristianstad
University
Sweden



Development and Evaluation of a Low-Power Ultrasonic Wake-Up System for Event-Driven Water Monitoring

Linn Hamadé & Sarah Didehvary

Relation to System Engineering course

System Engineering Project:

- Initial wake-up concept
- Hardware prototyping
- Camera trigger integration
- Basic event detection



Bachelor Thesis Contributions:

- Final system implementation
- Experimental evaluation
- Power consumption
- Detection performance
- Battery lifetime estimation

Waltero Context & Problem

→ Existing Waltero system

- Camera-based monitoring
- Captures images periodically
- Limited monitoring between captures

→ Problem

- Events occur infrequently
- Camera consumes significant power
- Continuous monitoring wastes energy



Waltero Camera-Based Monitoring Device

Waltero's Need



Example of a difficult-to-monitor drainage environment

No continuous monitoring



Late problem detection



Reactive maintenance



Higher maintenance costs

Proposed Event-Driven Monitoring Architecture

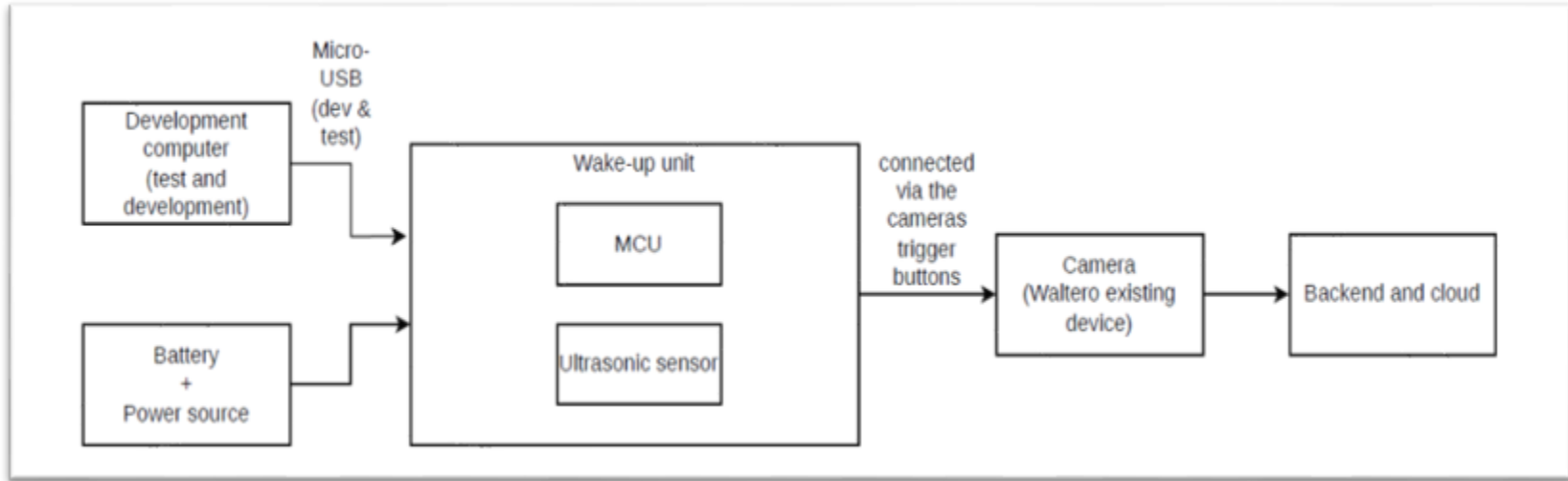
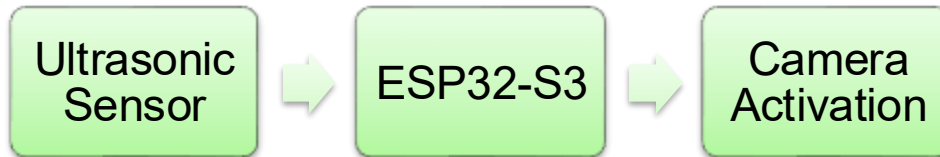


Figure 1. Proposed event-driven monitoring architecture integrating ultrasonic sensing, ESP32-S3 processing, and camera activation.



Research Questions & Objectives

RQ1

How reliably can an ultrasonic-based wake-up unit detect water-level events and trigger system activation?

RQ2

How does ultra-low-power operation affect detection accuracy, responsiveness, and overall system performance?

Research Objective

Balance reliability and energy efficiency

Literature Search Method

Search Databases

- HKR SuperSearch
- IEEE Xplore
- ScienceDirect

Selection Criteria

- 2022+
- Peer-reviewed
- Relevant to project
- Practical implementation



Example Refinement:

Embedded Monitoring: 40,279 → 7,909

Ultrasonic Monitoring: 1,074 → 40

Radar Comparison: 18,958 → 22

Literature Findings

Event-Driven Systems

- ✓ Lower power consumption
- ✓ Longer battery life
- ✗ Dependent on threshold selection

Ultrasonic Monitoring

- ✓ Non-contact sensing
- ✓ Low complexity
- ✗ Sensitive to reflections and disturbances

Radar Monitoring

- ✓ High accuracy
- ✓ Robust operation
- ✗ Higher system complexity

Critical Analysis of Existing Research

Limited Validation

- ✗ Many studies simulation-based
- ✗ Limited real-world testing

Application-Specific Solutions

- ✗ Designed for specific environments
- ✗ Difficult to generalize

System Integration

- ✗ Focus on individual components
- ✗ Few complete monitoring systems

Research Gap & Contribution

Existing Research

- ✓ Low-power monitoring
- ✓ Ultrasonic sensing
- ✓ Event-driven systems
- ✗ Limited integration
- ✗ Limited real-world evaluation

Our Contribution

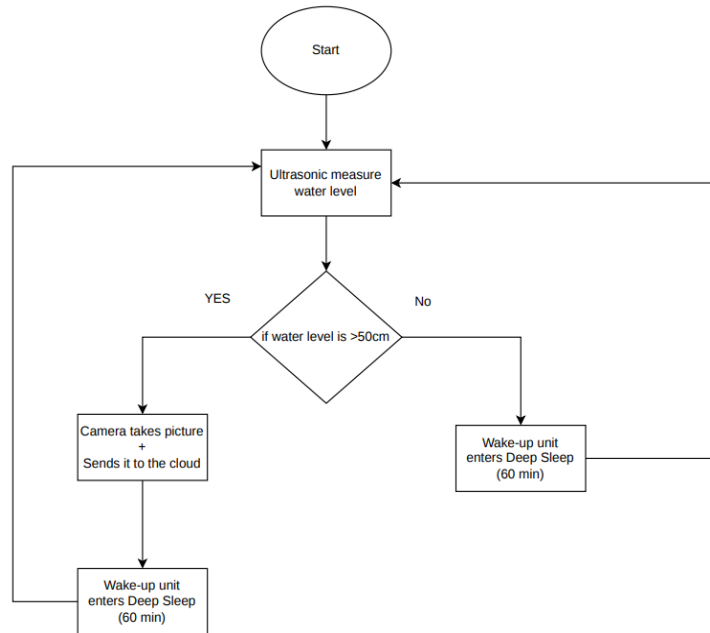
- ✓ ESP32-S3 wake-up unit
- ✓ Ultrasonic event detection
- ✓ Camera activation
- ✓ Experimental evaluation
- ✓ Battery lifetime analysis

Key Contribution:

An integrated low-power monitoring architecture combining sensing, event detection, and system evaluation.

Event-Driven Monitoring Workflow

Figure 2. Operational workflow of the proposed event-driven monitoring system.



How we tested → What we measured → What we found → What it means

Experiment setup

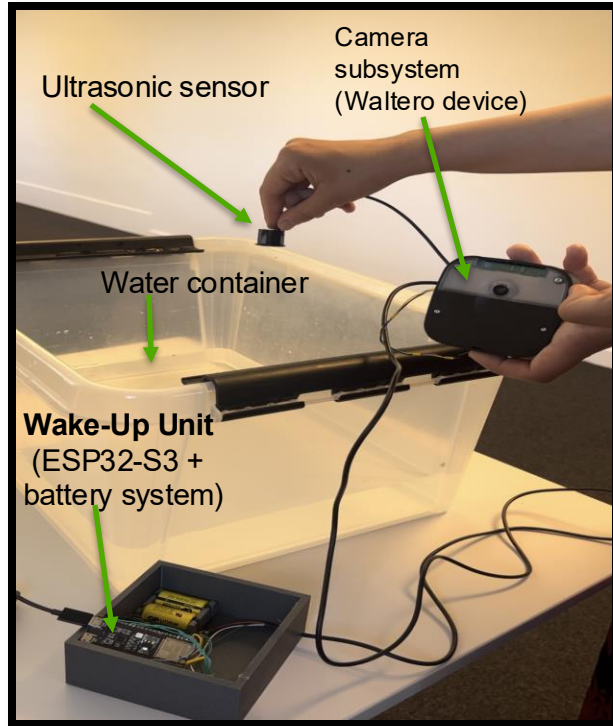


Fig 3. Controlled laboratory testing environment used for prototype evaluation.

Hardware Platform:

- ESP32-S3 DevKitC-1
- AJ-SR04M Ultrasonic Sensor
- Battery-powered Wake-Up Unit

Evaluation Focus:

- Power Consumption
- Wake-Up Behaviour
- Measurement Stability
- Event Detection Performance

Detection Test Design

Table 1. Detection test matrix used during experimental evaluation.

Distance (meter)	Still Water	Disturbe d Water	Reflectiv -e Surface	Obstructi -on
0.5	✓	✓	✓	✓
1.5	✓	✓	✓	✓
3.0	✓	✓	✓	✓

**3 Distances × 4 Conditions × 5 Repetitions = 60
Experimental Trials**

- **Still water** = Flat and undisturbed water surface
- **Disturbed water** = Water surface with manually created ripples
- **Nearby reflective surface** = Metal plate placed beside plastic container
- **Partial obstruction** = Cardboard sheet partially blocking sensing path

Power Consumption Results

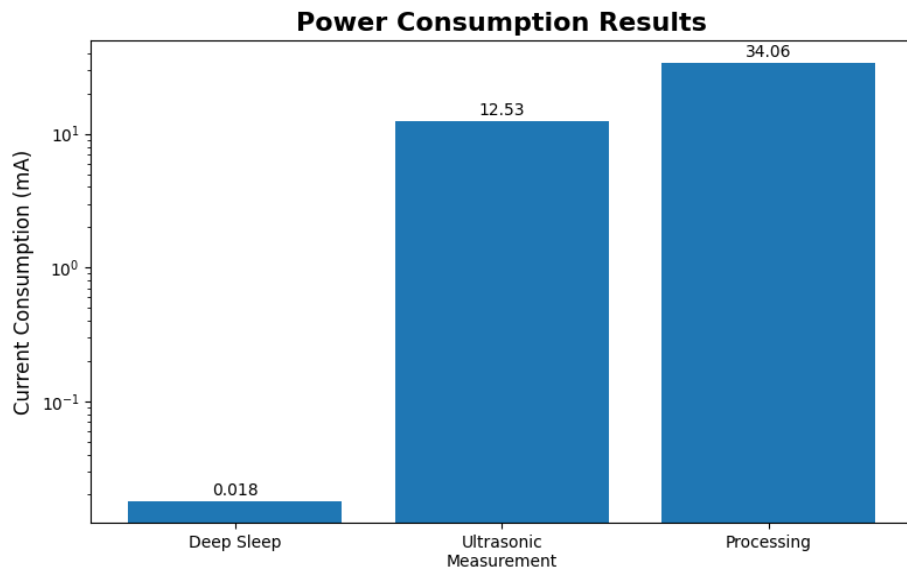


Figure 4. Measured current consumption for the different operating states of the system.

- **$\sim 1900 \times$ lower current consumption in deep sleep compared to processing mode**

Three operating states:

→ Deep Sleep → Measurement → Processing

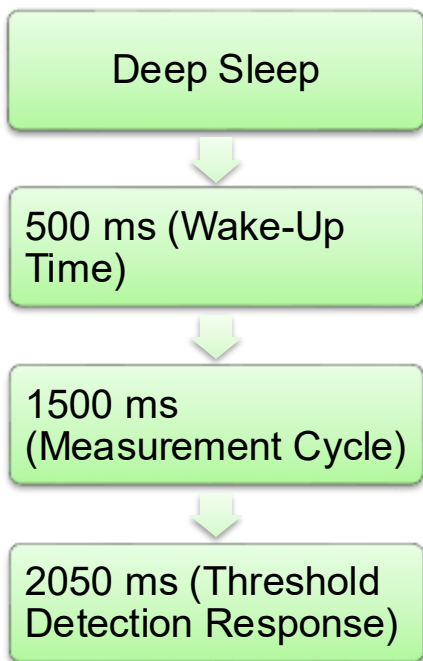
Key Finding:

- Deep Sleep: **0.018 mA**
- Measurement: **12.53 mA**
- Processing: **34.06 mA**

Note:
This excludes
camera power
consumption

Periodic deep-sleep
operation significantly
reduced energy
consumption.

Wake-Up Behavior



Findings of the wake-up behavior:

- Stable wake-up timing
- Fast response despite low-power operation
- Deep sleep maintained between measurement intervals

Figure 5. Wake-up and response-time measurements for the ESP32-S3 during periodic operation.

Ultrasonic Measurement Stability

Were the measurements stable?

Reliable Conditions:

- **Still Water**
 - ✓ Stable measurements
 - ✓ Event detected
- **Disturbed Water**
 - ✓ Minor variations
 - ✓ Event detected
- **Reflective Surface**
 - ✓ Minor variations
 - ✓ Event detected

Challenging Condition:

- **Partial Obstruction**
 - X Unstable readings
 - X Incorrect distance values
 - X Reduced detection reliability



Ultrasonic sensing
was reliable under
most
environmental
conditions.

Threshold Detection Performance

Did the system make the correct decision?

Successful Detection:

Table 2. Positive threshold-detection results under different environmental conditions.

Condition	Event Detected	False Trigger
Still water	YES	NO
Disturbed water	YES	NO
Reflective Surface	YES	NO

Detection Issues:

Table 3. Negative threshold-detection results under different environmental conditions.

Condition	Event Detected	False Trigger
Partial Obstruction	NO	YES

False Trigger: The system activated even though no valid threshold event had occurred.

Threshold-based detection worked well but was sensitive to signal obstruction.

Discussion

Table 4. Summarizes the discussion.

✓ Strengths	✗ Limitations
Event-driven operation = reduced unnecessary subsystem activity	Partial obstruction
periodic wake-up operation + deep sleep = Low-power performance	Environmental influences
Ultrasonic had mostly reliable detections	Controlled testing only
	No outdoor deployment

- Event-driven operation successfully reduced energy consumption while maintaining acceptable monitoring performance

Research Question Evaluation

RQ1: How reliably can an ultrasonic-based wake-up unit detect water-level-related events and trigger system activation under realistic monitoring conditions?

Findings:

Reliable event detection under:

- Still water conditions
- Disturbed water conditions
- Reflective surface conditions

Limitation:

- Partial obstruction reduced measurement reliability
- Incorrect readings caused detection errors

RQ1 Answer:

- Reliable under most tested conditions
- Sensitive to signal obstruction

RQ2: How does ultra-low-power operation affect detection accuracy, responsiveness, and overall system performance?

Findings:

- Deep-sleep current: 0.018 mA
- Response time: 2050 ms
- Stable operation during repeated wake-up cycles

RQ2 Answer:

- Significant power reduction
- Acceptable responsiveness
- Supports long-term operation

Conclusion & Future Work

Conclusions

- Reduced power consumption
- Reliable event detection
- Event-driven camera activation

Future Work

- Adaptive thresholds
- Signal filtering
- Wireless reporting
- Outdoor deployment

Event-Driven Monitoring = Lower Power + Reliable Detection

Acknowledgements

Supervisor:

Fredrik Frisk

- Guidance and continuous feedback throughout the thesis project

Company:

Waltero (Anders Carlberg and Viktor Andersson)

- Providing the practical monitoring use case
- Access to the camera-based monitoring system
- Technical insights and project support

Special Thanks

Supreet Yadav

Friend and student at Kristianstad University:

- Guidance and support throughout the thesis project
- Assistance with critical soldering of the wake-up unit during the System Engineering project
- Valuable technical discussions and feedback

Thank you!

